State of the Art and Related Work

Project: Heart Disease Prediction using Multilayer Perceptron

Juan Manuel Ruiz Llamas

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Abstract

This document presents a comprehensive review of the literature related to heart disease prediction using the UCI repository dataset. We analyze the performance of classical and neural algorithms found in the state of the art, with a specific focus on studies utilizing the combined dataset and Multilayer Perceptron (MLP) architectures. Finally, the internal mechanisms of two key proposals are dissected to provide a theoretical benchmark and methodological context for our proposed implementation.

1 Introduction

The prediction of heart disease using the UCI Machine Learning Repository dataset is a benchmark problem in the field of medical informatics. Since its publication by Detrano et al. in 1989 [1], the dataset has been extensively utilized to evaluate various supervised learning algorithms. This section summarizes relevant approaches, specifically focusing on Artificial Neural Networks (ANNs) and Multilayer Perceptrons (MLPs), to establish a comparative baseline for the development of our predictive model.

2 Approaches using the UCI Heart Disease Dataset

2.1 Classical Machine Learning Approaches

Early work on this dataset focused on probabilistic and statistical methods. Detrano et al. [1], the creators of the dataset, achieved an accuracy of approximately 77% using Logistic Regression, establishing the first baseline for the problem.

In later comparative studies, Palaniappan and Awang [3] developed an Intelligent Heart Disease Prediction System (IHDPS) comparing Naïve Bayes, Decision Trees, and Neural Networks. Their results indicated that Naïve Bayes reached 86.53%, outperforming Decision Trees. This suggests that the features in the dataset may exhibit strong conditional independence, a property that our MLP will need to capture through non-linear mapping rather than probabilistic assumptions.

Similarly, Shouman et al. [8] investigated K-Nearest Neighbors (KNN) and Decision Trees, finding that applying discretization to continuous variables significantly improved accuracy, reaching up to 84.1% with Decision Trees.

2.2 Multilayer Perceptron (MLP) and Deep Learning Approaches

Artificial Neural Networks have been a popular choice for this task due to their ability to model non-linear relationships between physiological risk factors.

Yan et al. (2006) [2] applied a Multilayer Perceptron to the dataset. Their study focused on optimizing the number of hidden layers and neurons. They reported that a standard MLP achieved an accuracy ranging widely from 63.6% to 82.3% depending on the sampling method. This high variance highlights the critical importance of robust validation strategies, such as K-Fold Cross-Validation, which we intend to implement to ensure the reliability of our results.

More recently, Latha and Jeeva (2019) [6] performed a comprehensive review. Their experiments showed that while a single MLP typically achieves approximately 82% accuracy, ensemble methods (bagging/boosting) could push performance slightly higher. This establishes a realistic performance ceiling for single-model architectures like the one proposed in our project.

It is crucial to note that many studies in the literature use only the "Cleveland" subset (303 samples) of the UCI dataset. Our project aims to utilize the combined dataset (Cleveland, Hungary, Switzerland, Long Beach VA), which contains more samples but also introduces more noise and missing values, making the classification task more challenging and realistic.

3 Detailed Case Study Analysis

To rigorously contextualize our proposed methodology, we selected two representative works. The first uses our exact data configuration (Combined Dataset), and the second uses our exact architecture (MLP) on an analogous medical problem (Diabetes).

3.1 Case 1: Use of Combined Dataset (Kahramanli & Allahverdi, 2008)

Most literature is limited to the "Cleveland" subset. However, since we aim to use the union of all four databases, the work of **Kahramanli and Allahverdi** [4] serves as a direct benchmark for this specific data configuration.

Approach: They proposed a hybrid system combining an Artificial Neural Network with Fuzzy Logic (*Fuzzy Neural Network*). The goal was to mitigate the uncertainty and noise inherent in the combined dataset, where the Switzerland and Hungary databases contain many missing values and inconsistencies.

Performance and Analysis:

- **Result:** They obtained an accuracy of **86.8%**.
- **Theoretical Implications for Our Project:** The success of their model is largely attributed to **Fuzzy Logic**. In a conventional MLP with Sigmoid/ReLU activation (our proposed architecture), decision boundaries are rigid. However, medical variables often have vague boundaries.
- **Expectations:** Kahramanli's hybrid approach likely represents an upper bound for performance on the combined dataset. Since our project implements a pure, non-hybrid MLP, we anticipate that the theoretical limit for our accuracy might be slightly lower than 86.8%. The difference will quantify how much "fuzzy" information is lost when forcing rigid decision boundaries on noisy medical data.

3.2 Case 2: MLP on Analogous Problem - Pima Indians Diabetes (Temurtas et al., 2009)

To evaluate the MLP architecture isolated from data quality issues, we analyzed the use of MLPs on the Pima Indians Diabetes dataset. This problem is formally nearly identical to ours: binary classification based on numerical physiological features.

Approach: Temurtas et al. [5] implemented a standard MLP to diagnose diabetes, focusing on the training algorithm. They compared traditional Stochastic Gradient Descent (SGD) against the **Levenberg-Marquardt (LM)** algorithm.

Performance and Analysis:

- **Result:** They achieved an accuracy of **82.37%** using 10-fold cross-validation.
- **Theoretical Implications for Our Project:**
  1. **Bayes Error:** The performance on Diabetes (approx. 82%) suggests an inherent "Bayes error" ceiling in tabular medical datasets with simple physiological features. This implies that regardless of network depth, there is a limit to the information contained in the features themselves. We should expect our Heart Disease model to converge within a similar performance band.
  2. **Scalability vs. Precision:** Temurtas proved that Levenberg-Marquardt (LM) often converges to better minima than SGD for small datasets. However, LM is computationally expensive ($O(N^3)$). Our proposed implementation utilizes SGD (or Adam variants). While SGD might require more careful hyperparameter tuning to match LM's precision, it provides a scalable foundation that is more aligned with modern Deep Learning practices for larger datasets.

4 Performance Summary

Table 1 summarizes the performance of various methods found in the literature. These values serve as the success criteria and benchmarks for our upcoming implementation.

Table 1: Summary of state of the art in heart disease prediction (Benchmarks).

Reference	**Technique**	**Dataset**	**Accuracy**
Detrano et al. (1989) [1]	Logistic Regression	Cleveland (303)	$\approx 77.0\%$
Yan et al. (2006) [2]	Standard MLP	Cleveland (303)	63.6% – 82.3%
Palaniappan et al. (2008) [3]	Naïve Bayes	Cleveland (303)	86.5%
Kahramanli et al. (2008) [4]	**Hybrid Neural Net**	**Combined (All)**	**86.8%**
Latha & Jeeva (2019) [6]	MLP (Backpropagation)	Cleveland (303)	82.0%

5 Conclusion

The literature review establishes clear expectations for the project. For the UCI Heart Disease dataset, classic MLP architectures typically range between 75% and 83% accuracy. Values exceeding 86% (like Kahramanli et al.) generally require hybrid architectures to manage the noise of the combined dataset.

Therefore, for our proposed *from-scratch* MLP implementation, achieving an accuracy consistent with the 78-83% band would constitute a successful validation. It would demonstrate that the model has effectively learned the non-linear mappings of the pathology within the limitations of a standard Feedforward architecture, matching the performance of established benchmarks in the field.

References

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